

TERMS OF REFERENCE
AI STACK – TECHNICAL SPECIFICATIONS
DATE: AUGUST 15, 2025

This document specifies the functional and technical requirements for the deployment of AI STACK - A Resource Management Platform to manage GPU/CPU/Memory resources, containerized workloads and AI computing services in a high-availability environment.

1. Project Application and Approval

- The platform shall provide project application approval, activation and cancellation functions.
- Upon project approval, GPU configuration parameters can be set.
- Applications within the configured range can proceed without administrator approval.

2. Project and User Management

- Administrators shall be able to create, modify and delete project content and members.
- The platform shall provide a batch deletion function for selected projects.
- Administrators shall be able to restrict specific GPU models for a project and project users shall only see the allowed GPU models.
- Multiple administrator roles shall be supported.
- User and administrator activity logs shall be recorded for all operations history on the platform.
- Administrators shall be able to adjust resource allocation (GPU/CPU/RAM quotas) for each project.

3. Image and Deployment Management

- The platform shall support automatic image pulling from NVIDIA GPU Cloud (NGC), allowing administrators to create service templates based on these images.
- NGC and Docker Hub image import functions shall be supported.
- Administrators shall be able to choose SSH access to containers via password or key-based authentication.
- User account integration with AD, LDAP, or SSO shall be supported.
- The platform shall provide a graphical dashboard for GPU resource allocation in two modes:
 - Aggregation Mode – support for one user using multiple GPU cards.
 - Shared Mode – support for multiple users sharing a single GPU card.

4. GPU Resource Partitioning and Cost Management

- GPU slicing capability shall be provided, allowing administrators to partition a single GPU's memory by model and define CPU and memory resources available for user selection.
- The platform shall support hourly GPU cost calculation.
- Per-project usage quotas shall be configurable, with options to increase quotas or set unlimited usage for specific projects.
- When a project exceeds its quota, the system shall issue a warning during container creation.

5. Monitoring and Reporting

- A centralized graphical dashboard shall monitor hardware utilization of all GPU servers (CPU, RAM, GPU, GPU RAM, etc.).
- The platform shall provide reservation-based resource scheduling, enabling users to book GPU/CPU/DRAM resources for specific time slots.

- If resources are unavailable, the system shall notify the user and display a waiting queue to the administrator.

6. CUDA Version Switching

- The platform shall provide a graphical interface for administrators to define and switch CUDA versions per node.

7. Storage and Security

- Support for MinIO object storage system integration.
- Scheduled container security scanning shall be available.
- Integration with Label Studio for data labeling functionality.
- The platform shall support management of both NVIDIA and AMD GPU resources.

8. High Availability (HA) and RCS(Rapid Container Service)

- HA design shall be supported to prevent single points of failure.
- Rapid container service (RCS) shall be provided, enabling users to deploy models/images without requiring GPU resources for inference.

9. Pre-installed AI Development Environment

- The platform shall include a complete deep learning development environment with:
 - Ubuntu 22.04 LTS or later
 - NVIDIA Driver
 - NVIDIA CUDA 11.5 or later
 - cuDNN
 - NVIDIA Docker
 - Jupyter
- During the warranty period, the OS and above software components shall receive continuous updates.

10. Elastic Training

- Elastic Training shall be supported, allowing flexible scaling of worker containers up or down during training tasks based on container cluster resource changes, improving resource utilization efficiency.

11. Licensing

- The platform's commercial software shall be provided under a perpetual license (buy-out model).
- The software license shall allow continued platform access after the initial 3-year license period.